CLYC Full-Energy Peak Efficiency for the DCS-Monitor

CLYC2 Distance Comparison, Peak Integration, Detector Resolution, and Flush-Source Response

DCS-Monitor simulation notes

November 3, 2026

Abstract

Here I compare simulated and measured full-energy peak efficiencies for the CLYC2 detector at nominal source-to-detector distances of 200 mm, 300 mm, and 400 mm. For the simulations, I ran isotropic Geant4 jobs with 10^7 gammas per point and integrated the full-energy peak in a $\pm 2\,\%$ total-energy window. For the measurements, I took the 2026-10-27 CLYC2 spectra, subtracted the 600 s environmental background, and fitted each peak with a Gaussian plus linear background over ± 1.5 FWHM (without using any external efficiency table for the distance comparison). At 661.64 keV the simulation/measurement ratio is around 0.98–1.03, and at 1332.51 keV it is 0.92–1.11. That being said, at 1173.23 keV the simulation comes out 19–36% higher than the measurement. Because the 1173.23 keV peak sits on top of the Compton continuum from the 1332.51 keV line, the Gaussian peak area has about a 10 % background-model uncertainty, however the observed excess is quite a bit larger than that: the measured 1173 keV/1332 keV ratio is 1.00–1.04, compared to 1.18 in the simulation. Since cascade summing is negligible at these distances, it seems that this line-dependent excess likely comes from the Co-60 source geometry or the detector response model. As a separate check, I also compared the 2 cm calibration file against a radioactive-decay simulation where the source puck is placed flush against the detector front face. This flush simulation matches reasonably well at 46.54 keV, 661.64 keV, and 1408.01 keV – as well as at 59.54 keV once pulse clustering is applied – while the 121.78 keV, 344.28 keV, and ^{60}Co lines still remain high. Fitting the 661.64 keV and 1332.51 keV lines gives a detector energy resolution of $R(E) = (123.9 \pm 0.4)/\sqrt{E/\mathrm{keV}}\,\%$ (i.e., FWHM $= 1.239\sqrt{E/\mathrm{keV}}$ keV).

1 Geometry and Solid Angle Efficiency Bound

In the Geant4 model (`GeometryCLYC.cc`), the detector uses a cylindrical CLYC crystal of diameter 25 mm and length 25 mm (radius $r = 12.5$ mm, density $3.31\,\mathrm{g\,cm^{-3}}$) inside a stepped aluminium casing. The front face of the crystal is recessed by 2 mm behind the outer front surface of the detector casing ($z_{\mathrm{recess}} = 2$ mm), with a 0.5 mm aluminium entrance window and a 1.5 mm Teflon reflector cap in between. Behind the 29 mm snout, the wider detector body and PMT are enclosed by a 0.64 mm-thick MuMetal magnetic shield. I modeled the calibration sources as isotropic point sources sitting at the centre of an epoxy puck of radius 10 mm, thickness 2 mm, and density $1.16\,\mathrm{g\,cm^{-3}}$ ($H_{32}C_{15}N_2O_4$).

For a given nominal source-to-detector distance d_{nom}, the actual distance from the source to the crystal front face is therefore $d = d_{\mathrm{nom}} + 2$ mm. Figure 1 shows the axial setup along with the solid-angle cone.

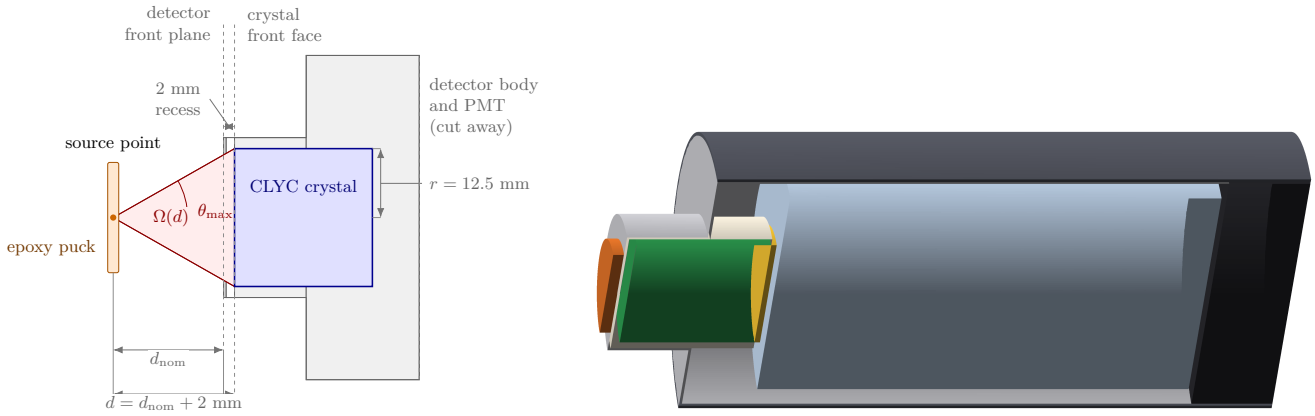
1.1 Solid angle and geometric acceptance

For an isotropic point source emitting into 4π, the solid angle subtended by the circular crystal entrance face of radius r at distance d along the detector axis is

$$\Omega(d) = 2\pi \left(1 - \frac{d}{\sqrt{d^2 + r^2}}\right), \tag{1}$$

so the corresponding geometric acceptance is

$$\epsilon_\Omega(d) = \frac{\Omega(d)}{4\pi} = \frac{1}{2}\left(1 - \frac{d}{\sqrt{d^2 + r^2}}\right) \times 100\%. \tag{2}$$



(a) Schematic axial section (not to scale).

(b) Geant4 cutaway half-section.

Figure 1: Axial geometry of the source and detector. Left: the point source is at the centre of the 2 mm epoxy puck; the detector front plane is at d_{nom} and the crystal front face is 2 mm farther, at $d = d_{\text{nom}} + 2$ mm. The red cone is the solid angle $\Omega(d)$ subtended by the crystal face of radius $r = 12.5$ mm. Right: Geant4 OGL cutaway view of the detector cut in half at $y = 0$, in the flush-source-to-detector-face geometry ($d_{\text{nom}} = 1$ mm, $d = 3$ mm); the source is at the left and the orange epoxy puck touches the detector front plane. The render uses opaque material colours: orange epoxy puck, grey aluminium casing, cream Teflon reflector, green CLYC crystal, yellow optical interface, pale blue PMT vacuum, dark PMT base, and dark grey MuMetal shield. The thin PMT glass envelope is omitted for clarity. The detector is 125 mm long. The left drawing is schematic; the distance scan at $d_{\text{nom}} = 200$ mm, 300 mm, and 400 mm is not drawn to scale.

For the source distances considered here, this gives:

$$\epsilon_\Omega = \begin{cases} 0.0955\,\%, & d_{\text{nom}} = 200\,\text{mm}\ (d = 202\,\text{mm}), \\ 0.0428\,\%, & d_{\text{nom}} = 300\,\text{mm}\ (d = 302\,\text{mm}), \\ 0.0242\,\%, & d_{\text{nom}} = 400\,\text{mm}\ (d = 402\,\text{mm}), \\ 38.33\,\%, & d_{\text{nom}} = 1\,\text{mm}\ (d = 3\,\text{mm}) \quad \text{(flush)}. \end{cases} \tag{3}$$

1.2 Maximum efficiency bound and intrinsic efficiency

We can factorize the total full-energy peak efficiency as

$$\epsilon = \epsilon_\Omega(d)\, \epsilon_{\text{int}}(E_\gamma, d), \tag{4}$$

where $\epsilon_{\text{int}} \in [0, 1]$ is the intrinsic efficiency – i.e., the probability that a gamma entering the crystal deposits all of its energy. Since $\epsilon_{\text{int}} \leq 1$, the geometric acceptance sets a strict upper bound on the efficiency:

$$\epsilon \leq \epsilon_{\max}(d) = \epsilon_\Omega(d). \tag{5}$$

2 Calculation Methods

2.1 Measured efficiency

Some notes about the experimental data and analysis: the 2026-10-27 CLYC2 spectra were converted from CNF to ROOT using `cnf2root`, which writes out the channel-count histogram, the calibrated energy histogram, and the acquisition metadata (live time and energy calibration). In the analysis I kept the native ~ 1.96 keV energy bins, since rebinning to 2 keV by bin centre would alias the two grids.

First, I subtracted the 600 s environmental background spectrum bin by bin. Of course, this removes room and cosmic background, however it does not remove the source-induced Compton continuum. I then fitted each full-energy peak with a Gaussian plus linear background over ± 1.5 FWHM around the observed peak (i.e., 1113.5 keV–1233.0 keV for the 1173.23 keV line, and 1264.4 keV–1400.6 keV for the 1332.51 keV line), taking the Gaussian integral as the measured peak area.

From this, the measured efficiency is given by

$$\epsilon_{\text{exp}} = \frac{N_{\text{peak}}}{A_{\text{exact}} P_\gamma t_{\text{live}}/100} \times 100\%, \tag{6}$$

where A_{exact} is the source activity on the measurement date, P_γ is the gamma-ray emission probability in percent, $t_{\text{live}} = 600\,\text{s}$, and N_{peak} is the Gaussian peak area (with its statistical uncertainty taken directly from the fit).

For the 1173.23 keV line, the peak sits on top of the Compton continuum from the 1332.51 keV line. As a cross-check, doing a local sideband integration over the ±1 FWHM peak region (with sidebands at 1.2–1.8 FWHM) gives a peak area that differs from the Gaussian-plus-linear fit by about 10 %. I treat this 10 % difference as the background-model systematic for the 1173.23 keV line. For the cleaner 661.64 keV and 1332.51 keV lines, the two methods agree to within about 1 %.

2.2 Simulated efficiency

The Geant4 simulations use `G4EmStandardPhysics_option4`. For each energy and distance point, I generated 10^7 monoenergetic gammas isotropically from the centre of the epoxy puck and summed all energy deposited in the CLYC crystal per event. Events are counted in the full-energy peak if they fall within a ±2% window:

$$0.98E_\gamma \leq E_{\text{dep}} \leq 1.02E_\gamma. \tag{7}$$

The simulated efficiency and its statistical uncertainty are then simply

$$\epsilon_{\text{sim}} = \frac{N_{\text{peak}}}{N_{\text{inc}}} \times 100\%, \qquad \sigma_{\text{sim}} = \frac{\sqrt{N_{\text{peak}}}}{N_{\text{inc}}} \times 100\%, \tag{8}$$

with $N_{\text{inc}} = 10^7$. Note that for this efficiency table I did not apply detector resolution smearing – in the monoenergetic simulation the full-energy peak is just a delta line, so the ±2% window already captures the entire peak.

To compare simulation and measurement, I use the ratio R and statistical pull:

$$R = \frac{\epsilon_{\text{sim}}}{\epsilon_{\text{exp}}}, \qquad \text{pull} = \frac{\epsilon_{\text{sim}} - \epsilon_{\text{exp}}}{\sqrt{\sigma_{\text{sim}}^2 + \sigma_{\text{exp}}^2}}. \tag{9}$$

2.3 Flush source response

In the flush geometry, the source puck sits directly against the detector front face. This places the point source at the puck centre 1 mm away from the detector front plane, and 3 mm from the crystal front face. Here I tracked the full decay chains using `G4RadioactiveDecayPhysics` with parent ions placed at rest at the puck centre ($N_{\text{decays}} = 10^6$ per source). In the default runs the long-decay threshold was set to 1×10^{60} y, and in the analysis I merged crystal hits whose leading-edge times are separated by at most the 1 µs CLYC resolving time. As a check on late decay-chain contributions, setting the threshold to 1×10^4 y stops the ^{241}Am chain at ^{237}Np and leaves the ^{137}Cs, ^{210}Pb, ^{152}Eu, and ^{60}Co results unchanged. The simulated peak efficiency is then calculated from the clustered pulse counts as

$$\epsilon_{\text{sim}}(E_\gamma) = \frac{N_{\text{pulse}}(E_\gamma)}{N_{\text{decays}} P_\gamma(E_\gamma)} \times 100\%, \tag{10}$$

with the statistical uncertainty taken from N_{pulse}.

2.4 The 2 cm calibration file

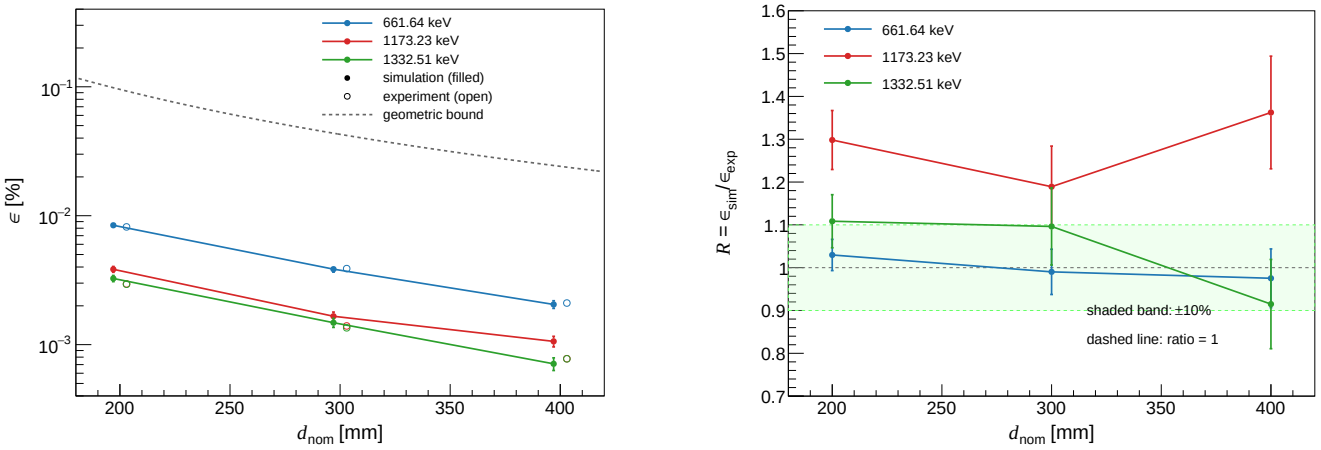
The existing calibration file `eff_clyc1.csv` is labelled as a 2 cm measurement. It does not have a detector column, so I treat it as CLYC1-only. Its printed numbers match Eq. (6) without an extra factor of 1000, and at lower energies (32 keV–121 keV) the efficiencies are quite high, ranging from 15.4 % to 23.9 %. Since those values are way above the geometric acceptance at 2 cm – and the file does not document the exact source reference point or detector identity – I compared them in Sec. 6 against a simulation with the source puck flush against the detector face.

3 Distance Comparison Results

Table 1 summarizes the nine CLYC2 distance-scan points, using the Gaussian peak areas from the 2026-10-27 spectra directly (with no external efficiency table). Figure 2 shows the resulting efficiencies and simulation/measurement ratios as a function of distance.

Table 1: Simulated and measured full-energy peak efficiencies for CLYC2 at the nominal source-to-detector distance d_{nom}. The source-to-crystal-front-face distance is $d = d_{\mathrm{nom}} + 2\,\mathrm{mm}$. The measured efficiency is from the Gaussian-plus-linear fit to the 2026-10-27 spectrum. All points satisfy $\epsilon < \epsilon_\Omega(d)$.

d_{nom} [mm]	E_γ [keV]	Source	ϵ_{sim} [%]	ϵ_{exp} [%]	R	Pull [σ]
200	661.64	^{137}Cs	0.008 43(29)	0.008 19(7)	1.030	+0.8
200	1173.23	^{60}Co	0.003 84(20)	0.002 96(3)	1.298	+4.4
200	1332.51	^{60}Co	0.003 26(18)	0.002 94(3)	1.108	+1.8
300	661.64	^{137}Cs	0.003 85(20)	0.003 89(5)	0.990	−0.2
300	1173.23	^{60}Co	0.001 66(13)	0.001 40(2)	1.189	+2.0
300	1332.51	^{60}Co	0.001 48(12)	0.001 35(2)	1.096	+1.1
400	661.64	^{137}Cs	0.002 05(14)	0.002 10(3)	0.975	−0.4
400	1173.23	^{60}Co	0.001 06(10)	0.000 78(2)	1.362	+2.8
400	1332.51	^{60}Co	0.000 71(8)	0.000 78(1)	0.915	−0.8



(a) Efficiency vs d_{nom} and the geometric bound.

(b) Simulation / measurement ratio.

Figure 2: Distance comparison for CLYC2. (a) Full-energy peak efficiency vs d_{nom}. Filled markers and solid lines are the simulation; open markers are the experimental Gaussian peak areas. The simulation and experiment markers are offset horizontally by ±3 mm so that they do not overlap; all quoted values and the table use the nominal d_{nom}. The dashed grey curve is the geometric bound from Eq. (2). (b) Ratio $R = \epsilon_{\mathrm{sim}}/\epsilon_{\mathrm{exp}}$. The green band is ±10 %; the dashed line is $R = 1$.

Now, some analysis of the distance scan:

1. For ^{137}Cs at 661.64 keV, the simulation/measurement ratio is 0.98–1.03 across the three distances, with pulls between −0.4 and +0.8σ.
2. For ^{60}Co at 1332.51 keV, the ratio is 0.92–1.11, with pulls between −0.8 and +1.8σ.
3. For ^{60}Co at 1173.23 keV, however, the ratio is 1.19–1.36, with pulls between +2.0 and +4.4σ.

The 1173.23 keV line is clearly the outlier here. In the data, the 1173 keV/1332 keV efficiency ratio is around 1.00–1.04, whereas the simulation predicts 1.18. Even accounting for the 10 % background-model systematic on the 1173.23 keV Gaussian area from the sideband check, that uncertainty does not explain away the discrepancy. And because the 1332.51 keV line agrees within 0.92–1.11, this is a line-dependent difference rather than a simple overall normalization offset.

I do not think this is coming from Co-60 cascade summing either: at 200 mm–400 mm, the total interaction probability per gamma in the crystal is only about 3.3×10^{-4}, so the probability for both cascade gammas to interact in the same decay is around 10^{-7} and the 2505.74 keV sum peak is negligible.

Instead, the 1173.23 keV excess seems to point to an energy-dependent effect in either the actual Co-60 source capsule/holder geometry, the active crystal volume, or the reflector/casing model – though with the current dataset alone it is hard to say which one is responsible.

4 Experimental Spectra and Peak Integration

As noted above, the 2026-10-27 CLYC2 runs are the only measured spectra used in this report. Following the CNF-to-ROOT conversion and 600 s environmental background subtraction, the ±1.5 FWHM Gaussian-plus-linear fits give the measured efficiencies listed in Table 1 (for example, the 1173.23 keV fit window is 1113.5 keV–1233.0 keV). Comparing those Gaussian areas against local sideband integration over the same peak region gives agreement to within about 1 % at 661.64 keV and 1332.51 keV, and about a 10 % difference at 1173.23 keV, which I use as the continuum-model systematic.

To look a bit more closely at the background shape under the Co-60 lines, Fig. 6 shows diagnostic Gaussian-plus-quadratic-background fits to the 200 mm Co-60 spectra. While Table 1 uses the linear background model for consistency across all runs, using a quadratic background here accounts a bit better for the curved Compton continuum from the 1332.51 keV line sitting underneath the 1173.23 keV peak. With the quadratic background, the fitted simulation/experiment peak-area ratios are 1.21 at 1173.23 keV and 1.04 at 1332.51 keV (compared to 1.30 and 1.11 in Table 1); the shift between the two fits reflects the background-model uncertainty on the 1173.23 keV peak.

5 Detector Resolution and Simulated Spectrum Comparison

5.1 Resolution model

From the same Gaussian-plus-linear fits at 200, 300, and 400 mm, I extracted the FWHM of the 661.64 keV, 1173.23 keV, and 1332.51 keV peaks. The widths are quite consistent across all three distances; Table 2 lists the mean values and the standard error of the mean. This gives an energy resolution $R = \mathrm{FWHM}/E$ of 4.80 % at 661.64 keV, 3.40 % at 1173.23 keV, and 3.41 % at 1332.51 keV. Because the 1173.23 keV width is biased a bit by the broad Compton background underneath the peak, I excluded it from the resolution fit and fitted only the two cleaner lines (661.64 keV and 1332.51 keV), which gives

$$\mathrm{FWHM}(E) = 1.239\,\sqrt{\mathrm{E/keV}}\,\mathrm{keV}, \qquad R(E) = \frac{123.9 \pm 0.4}{\sqrt{E/\mathrm{keV}}}\,\%.$$

The 1173.23 keV point sits 6.2 % below this curve, which is consistent with what I would expect from the background-model bias.

Table 2: Measured FWHM and resolution of the three CLYC2 lines, averaged over the 200, 300, and 400 mm spectra. The uncertainty is the standard error of the mean over the three distances. The model is $R(E) = 123.9/\sqrt{E/\mathrm{keV}}\,\%$, fitted to the 661.64 keV and 1332.51 keV lines; the 1173.23 keV point is a cross-check.

E_γ [keV]	n	FWHM [keV]	R [%]	R_{model} [%]	residual [%]	fit
661.64	3	31.786(86)	4.804(13)	4.818	−0.3	yes
1173.23	3	39.837(183)	3.396(16)	3.618	−6.2	no
1332.51	3	45.372(550)	3.405(41)	3.395	+0.3	yes

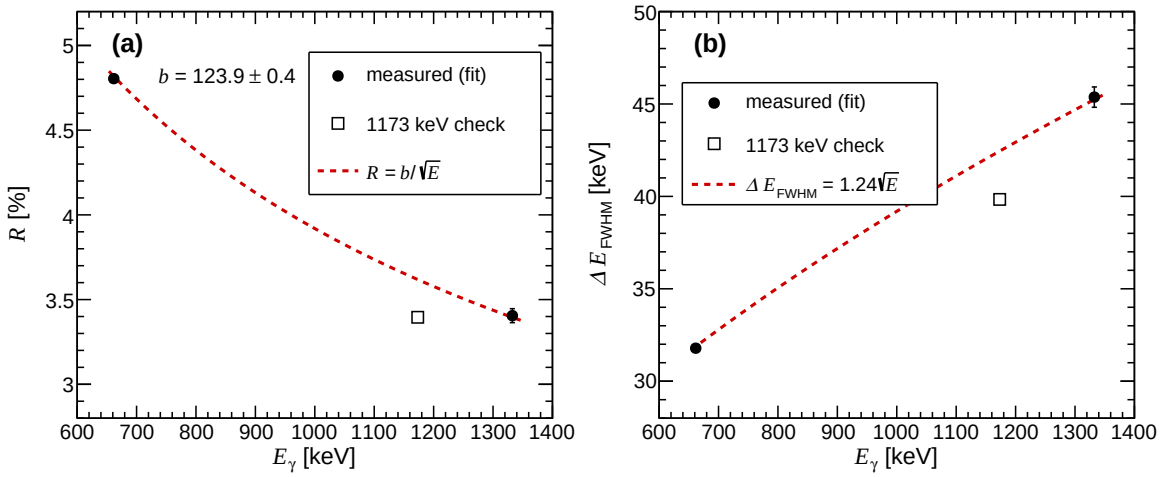


Figure 3: (a) Resolution $R = \mathrm{FWHM}/E$ and (b) FWHM versus E_γ. Filled circles are the 661.64 keV and 1332.51 keV lines used in the fit; the open square is the 1173.23 keV cross-check. The dashed curve is $R(E) = 123.9/\sqrt{E/\mathrm{keV}}\,\%$ ($\mathrm{FWHM} = 1.239\sqrt{E/\mathrm{keV}}\,\mathrm{keV}$).

I only use this resolution curve to smear the simulated spectra for visual/spectral comparison: for each event with deposited energy E_{dep}, the energy is drawn from a Gaussian with mean E_{dep} and standard deviation $\sigma(E_{\mathrm{dep}}) = \mathrm{FWHM}(E_{\mathrm{dep}})/2.3548$. This smearing conserves the total number of events and does not alter the integrated peak areas (of course, between and below the two fitted lines the curve is just an extrapolation).

5.2 Simulated and experimental spectra at 200 mm

Now to the plots comparing the full spectra at 200 mm. For these, I ran 10^7-event monoenergetic simulations with the GPS emission restricted to a 15 degree half-angle cone around the detector axis to get better statistics in the crystal. Restricting the emission to the cone boosts the number of gammas reaching the crystal by the inverse of the cone solid-angle fraction, $f = (1 - \cos 15^\circ)/2 = 0.017037$, corresponding to an effective isotropic incident count of $N_{\mathrm{inc,eff}} = N_{\mathrm{gen}}/f = 5.87 \times 10^8$. As a sanity check, the cone runs reproduce the isotropic full-energy peak efficiencies within about 1σ: 0.008 71 % vs. 0.008 43 % at 661.64 keV, 0.003 73 % vs. 0.003 84 % at 1173.23 keV, and 0.003 08 % vs. 0.003 26 % at 1332.51 keV. Since the 15 degree cone cut only drops wide-angle gammas that would have scattered into the crystal from outside the cone (which contribute mainly to the continuum), the full-energy peak comparison remains quantitative while the low-energy continuum comparison is more qualitative.

Both the smeared and unsmeared simulated spectra use the exact same events and scaling. With the expected number of emitted gammas given by

$$N_{\mathrm{emit}} = A_{\mathrm{exact}} P_\gamma t_{\mathrm{live}}/100,$$

the histogram scale factor is $N_{\mathrm{emit}} f/N_{\mathrm{gen}}$. For Cs-137 at 200 mm, $N_{\mathrm{emit}} = 2.834 \times 10^8$ and the scale factor is 0.4828. For Co-60, I summed the 1173.23 keV and 1332.51 keV runs, scaling each by its respective N_{emit} (7.031×10^8 and 7.032×10^8, scale factor 1.198). As before, the unsmeared histogram fills E_{dep} directly, while the smeared histogram draws from a Gaussian with $\sigma(E_{\mathrm{dep}}) = \mathrm{FWHM}(E_{\mathrm{dep}})/2.3548$. Since the two lines are generated in separate monoenergetic runs, the 2505.74 keV coincidence sum peak is not present in the simulation.

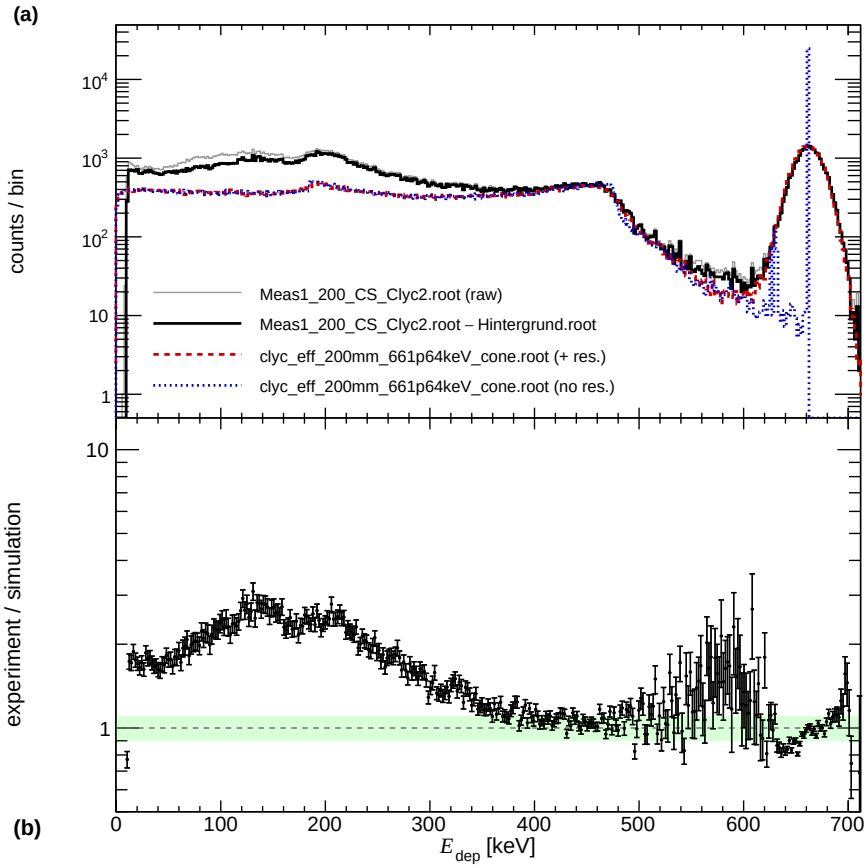


Figure 4: Simulated and experimental Cs-137 spectra at $d_{\mathrm{nom}} = 200\,\mathrm{mm}$. (a) Raw experimental spectrum (grey), 600 s background-subtracted spectrum (black), resolution-smeared simulation (red dashed), and simulation without resolution (blue dotted), all scaled to the measured live time and activity. (b) Ratio of the background-subtracted spectrum to the resolution-smeared simulation over the full x-range; the ratio axis is logarithmic, the dashed line is ratio = 1, and the green band is ±10%. The simulation uses the 15 degree cone run; the 661.64 keV peak is quantitative, the low-energy continuum is qualitative because the simulation contains only the 661.64 keV gamma. The experimental file is `Meas1_200_CS_Clyc2.root`, the background file is `Hintergrund.root`, and the simulation file is `clyc_eff_200mm_661p64keV_cone.root`.

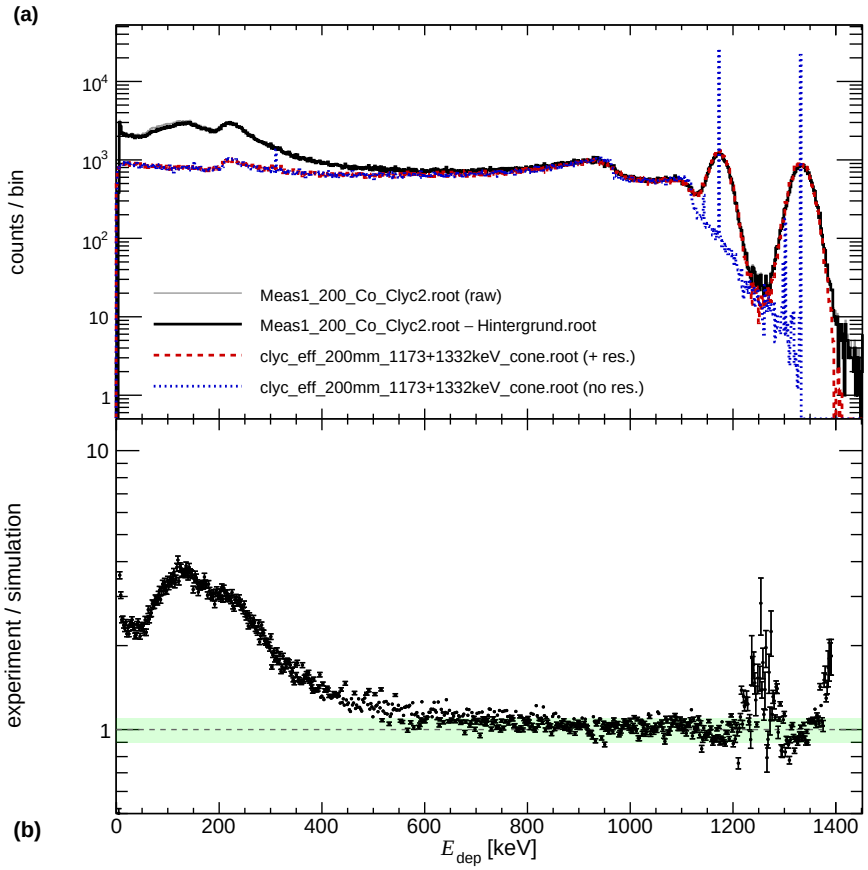


Figure 5: Simulated and experimental Co-60 spectra at $d_{\mathrm{nom}} = 200\,\mathrm{mm}$. (a) Raw experimental spectrum (grey), 600 s background-subtracted spectrum (black), and the sum of the 1173.23 keV and 1332.51 keV simulations with resolution (red dashed) and without resolution (blue dotted), all scaled to the measured live time and activity. (b) Ratio of the background-subtracted spectrum to the resolution-smeared simulation over the full x-range; the ratio axis is logarithmic, the dashed line is ratio = 1, and the green band is ±10%. The 2505.74 keV sum peak is absent from the monoenergetic simulation. The ratio in (b) is a bin-by-bin interval comparison; at 1173.23 keV the interval contains Compton continuum from the 1332.51 keV line, so it is not the full-energy peak efficiency of Table 1. The experimental file is `Meas1_200_Co_Clyc2.root`, the background file is `Hintergrund.root`, and the simulation is the sum of `clyc_eff_200mm_1173p23keV_cone.root` and `clyc_eff_200mm_1332p51keV_cone.root`.

For Cs-137 at 200 mm (Fig. 4), the resolution-smeared simulation matches the 661.64 keV peak quite well: in the ±2% energy window, the ratio of background-subtracted data counts to the smeared simulation is 0.955, and the smeared simulated peak area is only about 5 % above the measured Gaussian area. At lower energies the simulation sits below the measured continuum, which is expected since the experimental spectrum includes room scatter and environment-scattered radiation that are not in the monoenergetic model. Also, the measured 200 mm Cs-137 spectrum does not show a resolved 32 keV Ba X-ray peak – likely because the source encapsulation absorbs those X-rays, whereas the monoenergetic run only generates the 661.64 keV gamma anyway. So the 661.64 keV peak itself is the main quantitative benchmark here.

For Co-60 at 200 mm (Fig. 5), Table 1 gave $\epsilon_{\mathrm{sim}}/\epsilon_{\mathrm{exp}} = 1.30$ at 1173.23 keV and 1.11 at 1332.51 keV. Note that the bin-by-bin ratio plotted in Fig. 5(b) looks a bit different near 1173.23 keV because it is an interval ratio rather than a background-subtracted peak-area ratio: the ±2% window at 1173.23 keV is 46.9 keV wide (1.18 FWHM) and includes Compton continuum from the 1332.51 keV line. Specifically, the background-subtracted data integral in that window is 23 396 counts (vs. 20 800 in the Gaussian peak), whereas the smeared simulation has 24 306 counts in the window (and about 26 200 in the full-energy peak). Taking the raw window ratio gives 0.963, whereas comparing the actual full-energy peak counts gives 20,819/26,210 = 0.79 (i.e., a simulation/measurement ratio of 1.26, consistent with the 1.30 in Table 1). Figure 6 shows the explicit peak-plus-background decompositions, where the fitted simulation/experiment peak ratios are 1.21 and 1.04. Even with the 10 % background-model uncertainty from Sec. 4, the 1173.23 keV simulation still sits high, while the 1332.51 keV line agrees within the peak-integration uncertainty. A couple of other small features in the measured Co-60 spectrum are worth

noting: there are weak lines at 141.8 keV and 222.1 keV that are not from Co-60 (and are absent in the simulation), and the 2505.74 keV sum peak is absent in the monoenergetic simulation as well. That being said, at 200 mm the sum peak is expected to be around 5 counts (with a summing-out probability below 0.05 % at 200–400 mm given the 3.3×10^{-4} single-gamma interaction probability), so cascade summing cannot explain the 1173.23 keV excess.

For reference, the conversion, analysis, and plotting scripts live in `report/eff_clyc/scripts/convert_exp_spectra.sh`, `analyze_exp_spectra.C`, `make_exp_sim_figs.C`, and `make_distance_figs.C`. The extracted peak-integration tables are in `report/eff_clyc/data/`, and the 15 degree cone simulations were run with `scripts/run_clyc_eff_cone_200mm.sh`.

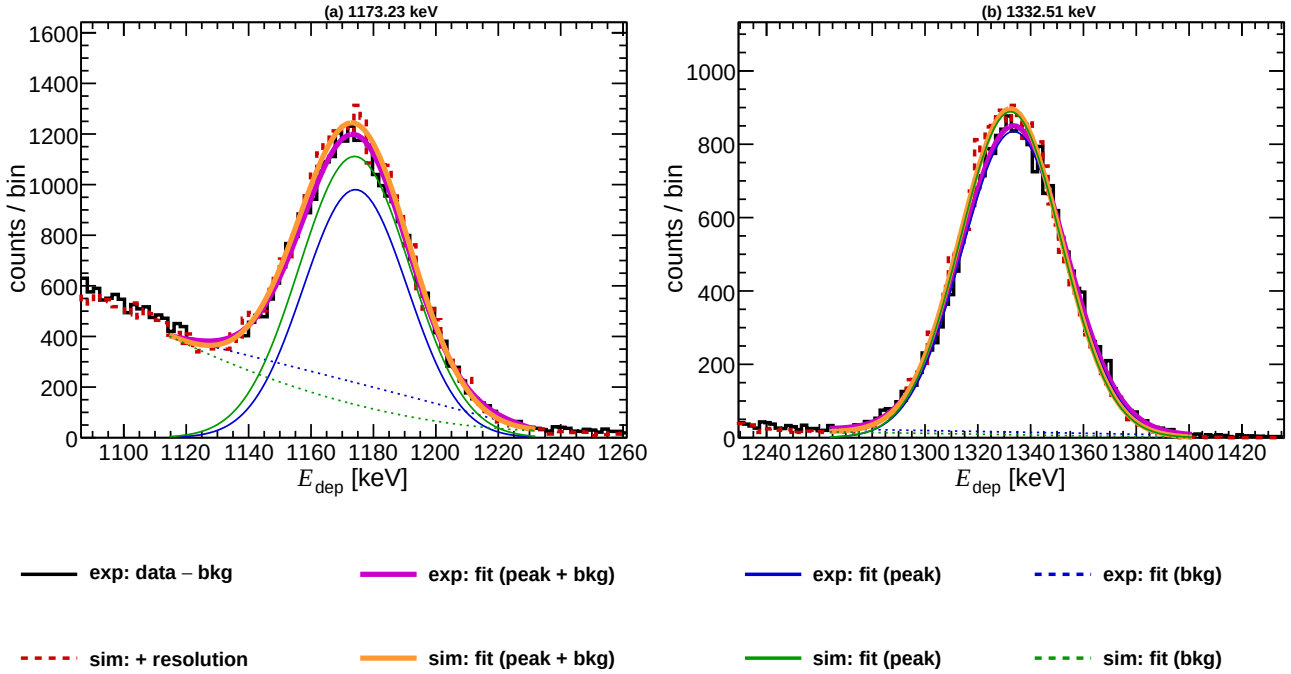


Figure 6: Gaussian-plus-background fits to the 200 mm Co-60 spectra. (a) 1173.23 keV and (b) 1332.51 keV. Black: experiment after the 600 s environmental background subtraction. Red dashed: resolution-smeared simulation. Both spectra are fitted over ±1.5 FWHM of the measured peak with a Gaussian plus quadratic background. For each fit, the thick magenta/orange curve is the peak-plus-background fit, the thin blue/green curve is the Gaussian peak, and the dotted blue/green curve is the background. The fitted peak areas are 20 592 (experiment) and 24 885 (simulation) at 1173.23 keV, and 20 670 and 21 438 at 1332.51 keV; the simulation/experiment ratios are 1.21 and 1.04. The quadratic background is used because the 1173.23 keV peak sits on the curved Compton continuum from the 1332.51 keV line. The data files are `Meas1_200_Co_Clyc2.root` and `Hintergrund.root`; the simulation is the sum of `clyc_eff_200mm_1173p23keV_cone.root` and `clyc_eff_200mm_1332p51keV_cone.root`.

6 Radioactive-Decay Source Response in the Flush Geometry

Table 3 summarizes the simulated full-energy peak efficiencies for the nine calibration lines when the source puck is placed flush against the detector front face ($d_{\mathrm{nom}} = 1$ mm, $d = 3$ mm; $N_{\mathrm{decays}} = 10^6$ per source). As mentioned earlier, `eff_clyc1.csv` has no detector column and is treated as CLYC1-only, though the simulated curve is identical for CLYC1 and CLYC2 since they share the same geometry. Figure 7 compares the flush simulation against the printed CSV values and the geometric acceptance bound, with the corresponding intrinsic efficiencies shown in panel (b).

Table 3: Simulated peak efficiencies in the flush-source-to-detector-face geometry ($d_{\mathrm{nom}} = 1\,\mathrm{mm}$, $d = 3\,\mathrm{mm}$; $N_{\mathrm{decays}} = 10^6$ per source). The efficiency is the number of 1 µs pulses in the ±2 % window divided by $N_{\mathrm{decays}} P_\gamma$. The source runs use the 1×10^{60} y threshold; the ^{241}Am 1×10^4 y cross-check is discussed in the text. The last column is $R = \epsilon_{\mathrm{flush}}/\epsilon_{\mathrm{CSV}}$.

Source	E_γ [keV]	$\epsilon_{\mathrm{flush}}$ [%]	ϵ_{CSV} [%]	R
^{137}Cs	32.00	15.900	18.50	0.859
^{210}Pb	46.54	25.233	23.85	1.058
^{241}Am	59.54	28.614	23.80	1.202
^{152}Eu	121.78	18.006	15.40	1.169
^{152}Eu	344.28	6.274	4.87	1.288
^{137}Cs	661.64	2.057	1.87	1.100
^{60}Co	1173.23	0.892	0.51	1.749
^{60}Co	1332.51	0.706	0.56	1.261
^{152}Eu	1408.01	0.431	0.40	1.079

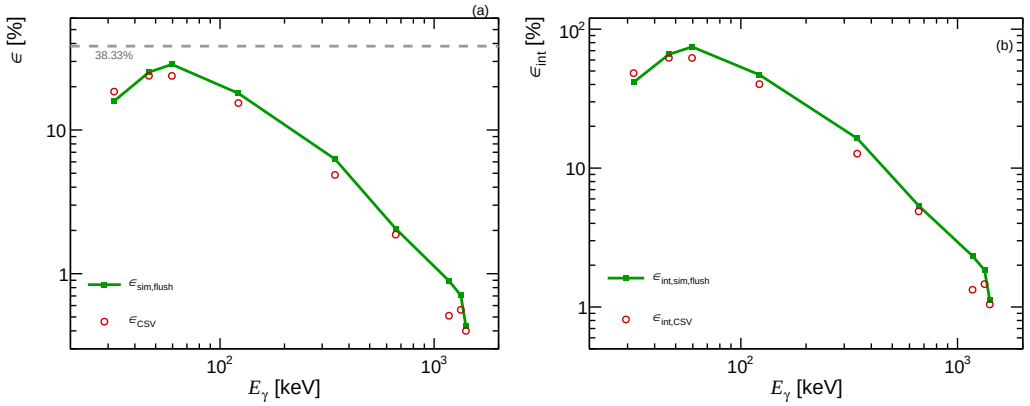


Figure 7: Simulated ϵ in the flush-source-to-detector-face geometry compared with the printed CSV values. (a) Raw efficiency; the dashed line marks the geometric acceptance $\epsilon_\Omega(3\,\mathrm{mm}) = 38.33\,\%$. (b) Intrinsic efficiency; the flush simulation and the CSV are divided by $\epsilon_\Omega(3\,\mathrm{mm})$.

Overall, the flush simulation behaves as expected physically: every point stays below the 38.33 % geometric ceiling, and the 32 keV efficiency (15.90 %) drops below the 46.54 keV and 59.54 keV points because photoelectric absorption inside the 2 mm epoxy puck attenuates the 32 keV X-ray. Above 122 keV, the efficiency falls off smoothly with energy down to 0.431 % at 1408 keV.

Just to recap the geometry here: placing the 2 mm-thick puck flush against the detector front face puts the source (at the puck centre) 1 mm in front of the casing surface, and since the crystal face is recessed by 2 mm, the source-to-crystal distance is $d = 3\,\mathrm{mm}$. Plugging $d = 3\,\mathrm{mm}$ into Eq. (2) gives $\epsilon_\Omega(3\,\mathrm{mm}) = 38.33\,\%$. Comparing the flush simulation to the printed CSV numbers in Fig. 7 and Table 3:

1. With the 1 µs pulse clustering applied, the flush simulation agrees reasonably well with the CSV at 46.54 keV, 661.64 keV, 1408.01 keV, and 59.54 keV ($R = 1.058$, 1.100, 1.079, and 1.202).
2. At 32.00 keV, the simulation is a bit lower than the CSV (15.90 % vs. 18.50 %, $R = 0.859$).
3. The 121.78 keV, 344.28 keV, and especially the ^{60}Co lines remain noticeably higher in the simulation ($R = 1.749$ and 1.261 for ^{60}Co).

Some notes on how the pulse clustering affects the 59.54 keV ^{241}Am line, which changed the most when moving from an event-total sum to pulse counting:

1. If I simply sum all hits per Geant4 event across the full decay chain, the apparent 59.54 keV efficiency in the flush geometry is only 11.41 %.
2. However, the `hits` tree actually stores one row per shower pulse. When I instead merge only rows whose leading-edge times are within 1 µs of each other, the efficiency jumps to 28.61 % for the 1×10^{60} y threshold run, and 28.06 % for the 1×10^4 y cross-check run.
3. That 0.56-percentage-point difference between the two threshold runs comes from late-chain decays: in the 1×10^{60} y file, 2204 out of the 101 387 rows in the 59.54 keV window occur at $t \geq 1 \times 10^{21}$ ns,

meaning they come from ^{237}Np or its daughters rather than the prompt ^{241}Am decay. Because those daughter decays are separated in time from the prompt pulse, time clustering keeps them from being summed into the prompt pulse (which is what caused the artificial drop to 11.41 % in the event sum), though in the 1×10^{60} y run they still get counted as separate pulses. So the 1×10^{60} y run is really a full-chain stress test, while the 1×10^{4} y result (28.06 %) is the physical value for a fresh ^{241}Am source.

Even with pulse clustering and full decay cascades included, the ^{60}Co lines in the flush simulation remain high compared to the CSV (0.892 % and 0.706 % vs. 0.51 % and 0.56 %). Because the flush ^{60}Co run simulates the full radioactive decay, coincidence summing-out of the 1173.23 keV and 1332.51 keV lines into the 2505.74 keV sum peak is already included (and is quite strong at $d = 3$ mm). So the takeaway seems to be that while a flush source placement makes the low- and high-energy anchor lines (46.54 keV, 59.54 keV, 661.64 keV, 1408.01 keV) plausible without tuning any parameters, source distance alone does not resolve the discrepancy on the ^{60}Co and ^{152}Eu intermediate lines.

7 Conclusions

1. The geometric solid angle sets a strict upper bound $\epsilon \leq \epsilon_\Omega(d)$ on the full-energy peak efficiency: 0.0955 % at 200 mm, 0.0428 % at 300 mm, 0.0242 % at 400 mm, and 38.33 % for the flush-source-to-detector-face geometry ($d = 3$ mm).
2. For ^{137}Cs at 661.64 keV, the simulation/measurement ratio across 200–400 mm is 0.98–1.03 (pulls between -0.4 and $+0.8\sigma$).
3. For ^{60}Co, the 1332.51 keV simulation/measurement ratio is 0.92–1.11 (pulls between -0.8 and $+1.8\sigma$), whereas the 1173.23 keV ratio is 1.19–1.36 (pulls between $+2.0$ and $+4.4\sigma$). The measured 1173 keV/1332 keV efficiency ratio is 1.00–1.04, compared to 1.18 in the simulation. Since cascade summing is negligible at 200–400 mm, this line-dependent excess likely points to an issue in the Co-60 source geometry or the detector response model.
4. All measured distance-scan efficiencies come directly from Gaussian-plus-linear fits to the 2026-10-27 CLYC2 spectra (no external efficiency table is used). Comparing the Gaussian fits against local sideband integration gives agreement within about 1 % at 661.64 keV and 1332.51 keV, and a 10 % background-model systematic at 1173.23 keV.
5. Comparing the 2 cm calibration file (`eff_clyc1.csv`) against the flush-source simulation ($d_{\text{nom}} = 1$ mm, $d = 3$ mm) gives reasonable agreement at 46.54 keV, 661.64 keV, 1408.01 keV, and – once 1 µs pulse clustering is applied – at 59.54 keV. That being said, the 121.78 keV, 344.28 keV, and ^{60}Co lines still sit high in the simulation, so an unrecorded flush source position alone does not explain all of the differences in that table.
6. Fitting the 661.64 keV and 1332.51 keV peak widths across 200–400 mm gives a detector resolution model of $R(E) = (123.9 \pm 0.4)/\sqrt{E/\text{keV}}$ % (i.e., FWHM $= 1.239\sqrt{E/\text{keV}}$ keV). Smearing the 200 mm simulated spectra with this resolution reproduces the measured 661.64 keV and 1332.51 keV peaks within about 5 %, while the 1173.23 keV full-energy peak area remains 19–36% above the measurement.